LiZARD

Liver Zonation Analysis, Robustness & Donor effects.

A donor-aware computational reanalysis of liver zonation in chronic disease.

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Abstract. Gribben et al. (Nature 2024) reported that, during human metabolic-dysfunction–associated steatotic liver disease (MASLD), hepatocytes progressively lose their porto-central *zonation* and acquire a cholangiocyte-like (biliary) identity. Using the deposited single-nucleus RNA-seq data, we asked whether the transcriptomic evidence for *progressive* de-zonation survives once the way each sample was obtained is taken into account. In this cohort, disease stage is collinear with tissue acquisition: the F0–F4 disease spectrum consists of 16-gauge needle biopsies, whereas the healthy and end-stage groups are deceased-donor whole-organ tissue, so the apparent healthy-to-end-stage trajectory is also an acquisition trajectory. The endpoint groups also carry a cross-lineage stress signature compatible with acute acquisition or handling effects. Restricting the analysis to the acquisition-matched needle biopsies (38 donors across F0–F4; 36 across the inferential F1–F4 axis) and working on raw, depth-normalized UMI counts with the donor as the unit of inference, we find **no clear transcriptional de-zonation signal**: selected pericentral, periportal, strict dual-coexpression and double-negative measurements are flat or non-monotone across fibrosis. Precision is limited: the F1-to-F4 pericentral-fraction difference is +3.45 percentage points (90% CI −13.63 to +20.54), and equivalence is not established at either the pre-specified ±5- or ±10-point margin. A genome-wide per-gene pseudobulk scan finds no zonation gene moving (GLUL F4-vs-F1 FDR 0.80); its only per-gene signal is a biliary/ductular-marker rise at cirrhotic F4 inside hepatocyte-labelled nuclei, whose source remains unresolved. Pre-specified gene-set tests (CAMERA, ROAST, GSEA) plus an independently computed per-cell measure identify one biopsy-internal trend: the **pericentral detox program dims**. The robust signal is the cytochrome-P450 xenobiotic core — the only set significant in both a competitive and a self-contained gene-set test (camera FDR 2×10^{-6}, roast FDR 0.015) — and an eight-draw, depth-matched per-cell summary declines from 11.86 to 9.02 transcripts per pericentral nucleus from F1 to F4 (donor-level Spearman $\rho = -0.461$, $p = 0.0041$). This is a *relative* decline within a fixed sequencing budget, not proven absolute molecule loss, and the data do not establish whether it reflects within-cell regulation or a shift among pericentral subzones. We re-examine only the single-nucleus transcriptional evidence; the paper's imaging, protein, organoid and spatial findings are not addressed.

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Supplementary report (optional). The required deliverable for this project is the presentation; this document is a complete written companion to it.

Course	Computational Genomics 76553 – Hackathon
Date	2026-06-23
Dataset	Gribben et al., *Nature* 2024, "Acquisition of epithelial plasticity in human chronic liver disease" — GSE202379 (single-nucleus RNA-seq component only)

1 Introduction

1.1 Hepatocyte zonation and epithelial plasticity

The liver's repeating functional unit, the lobule, carries a steep chemical gradient: blood enters at the portal triads, relatively oxygen- and nutrient-rich, and flows inward to the central vein, where oxygen is depleted. Because a hepatocyte's environment depends on where it sits along this porto-central axis, hepatocytes specialise smoothly along it — a division of labour called **zonation**. Broadly, the **periportal** pole runs much of the urea cycle (ASS1) and gluconeogenesis (PCK1, ALDOB), while the **pericentral** pole runs glutamine synthesis (GLUL) and cytochrome-P450 xenobiotic metabolism (CYP3A4, CYP2E1), maintained by an oxygen and Wnt/R-spondin gradient emanating from central-vein endothelium. Zonation is *intra*-hepatocyte variation — a question of *where on the axis* a hepatocyte sits.

This must be kept distinct from **epithelial plasticity**, which is variation *between* cell types: the normally locked boundary between hepatocyte (ALB) and cholangiocyte/biliary (KRT7, KRT19, SOX9) identity blurring to produce biphenotypic cells. De-zonation is a hepatocyte forgetting its *address*; plasticity is a hepatocyte drifting away from *being a hepatocyte*. The two phenomena are measured with separate gene sets and must not be conflated.

1.2 The paper and the motivation

Gribben et al. (*Nature* 2024) profiled 47 liver samples by snRNA-seq across the MASLD spectrum (healthy, NAFLD/MASLD, MASH, cirrhosis, end-stage), yielding ∼99,809 QC-passing nuclei including 69,426 hepatocytes. Their headline biology is hepatocyte→cholangiocyte transdifferentiation

driven by PI3K-AKT-mTOR, supported by organoid and imaging evidence; as a secondary observation they report that hepatocytes progressively lose zonation, evidenced in the snRNA data by a pericentral-vs-periportal marker correlation (with a Welch's-t comparison) pooled across donors and by a stage-split hepatocyte embedding. It is a careful, high-quality study, and the de-zonation reading is intuitive. The motivation here is narrow and specific: the snRNA de-zonation statistic pools the entire cohort across a sampling discontinuity, and we wanted to know whether the transcriptional signal survives when that discontinuity is removed.

1.3 Research question

- **Primary.** Across an *acquisition-matched* MASLD axis (needle-biopsy fibrosis F1→cirrhotic F4), is there a transcriptional de-zonation signal — does hepatocyte zonation progressively collapse as the paper reports — or does that signal disappear once acquisition is matched?
- **Secondary.** Given no detectable de-zonation signal, what *else* changes genome-wide in the hepatocyte transcriptome across the matched disease axis — and what is the source of any such change?

1.4 Hypothesis and operational goals

The original project was framed around three hypotheses (H1: a de-zonation score rises monotonically with stage; H2: zone-specific programs — e.g. pericentral CYP detox — collapse earlier/faster; H3: the most de-zonated hepatocytes are the ones gaining biliary identity). Our operational goals reframe these into testable count-based endpoints that survive the confound: (i) quantify the tissue-acquisition confound directly; (ii) replace the relative, depth-sensitive marker-correlation "ruler" with a mechanism-resolving anchor classification on raw counts; (iii) bound how large a zonation shift the matched axis could have hidden; and (iv) run an honest genome-wide discovery scan for any other hepatocyte program.

Thesis. The headline transcriptional de-zonation pattern in this cohort is inseparable from tissue acquisition: the healthy and end-stage groups are deceased-donor organ tissue, whereas the disease spectrum is sampled by needle biopsy. On the acquisition-matched biopsy axis, selected zonation measurements show **no clear progressive de-zonation signal**, but the available donor counts do not establish tight equivalence at ± 5 or ± 10 percentage points. The per-gene genome-wide scan likewise finds no zonation gene moving; its only per-gene signal is a biliary-marker rise whose source remains an open compositional-versus-intrinsic question. One biopsy-internal trend is supported by gene-set and per-cell testing: the **pericentral detox program dims** in relative abundance across fibrosis. The robust part is the cytochrome-P450 xenobiotic core, significant in both a competitive and a self-contained gene-set test and accompanied by a repeat-averaged per-cell decline. We re-examine only the single-nucleus transcriptional leg of the paper; its imaging, protein, organoid and spatial-architecture evidence are not addressed.

2 Data and methods

2.1 Dataset and the matched axis

The data are the snRNA-seq component of GSE202379: 47 donors, ∼99,809 QC-passing nuclei, 69,426 of them hepatocytes. From Paper 1's own Methods, tissue was obtained two different ways. The **disease spectrum** (38 donors, fibrosis F0–F4 = 2/8/12/12/4 donors) came from *16-gauge end-cut needle core biopsies* ("two ultrasound-guided needle core liver biopsies of approximately 2 cm"), right-lobe by construction. The **healthy** group (4 donors) and the **end-stage** group (5 donors) are *deceased-donor whole-organ cubes* ("a cube of approximately 1 cm^3 was cut and frozen"), and for the explants (and two healthy donors) tissue was sampled from all three lobes (left, right, caudate). Our disease axis is therefore the **needle-biopsy donors only**.

A scoping note carried throughout: our "biopsy F4" (n=4) is **NASH-with-cirrhosis needle biopsies**, distinct from the 5 end-stage *explant* donors, although both carry fibrosis score F4. The matched axis thus tests *into established cirrhosis*, which is where de-zonation should be most visible if it occurs. F0 (n=2) is shown descriptively but is too thin to anchor inference; the inferential weight rests on F1–F4, with the best-powered interior at F1–F3 (n=8/12/12).

2.2 Raw counts, not SCT

The shipped expression matrix is an SCT (regularized-negative-binomial) model output, with depth squeezed to a synthetic budget — appropriate for visualization but not for molecule-level questions such as co-detection of two markers in one nucleus or sensitivity to ambient RNA. All quantitative claims here are computed on the **raw RNA UMI counts** extracted from the RNA assay of the Seurat object, never on the SCT matrix.

2.3 Depth normalization by binomial down-thinning

Deeper-sequenced nuclei detect more genes for purely technical reasons, which would inflate any detection- or co-detection-based metric. To make detection fair across nuclei we **down-thin** each nucleus's raw UMIs to a common budget of **1,500 UMIs** by binomial thinning (nuclei below the budget are dropped), and average the resulting metrics over Monte-Carlo draws. The headline endpoints were re-checked at budgets of 1,000 / 1,500 / 3,000 UMIs; Monte-Carlo SD on donor-median fractions was 0.006–0.010.

2.4 Procurement-stress module

To quantify acute handling/dissociation stress we summed raw UMIs of an 8-gene module of immediate-early and heat-shock genes: **FOS, JUN, JUNB, JUND, ATF3, DUSP1, HSPA1A, HSPA1B**. A separate HIF/hypoxia panel (VEGFA, SLC2A1, LDHA, CA9, ENO1, PGK1) was held out as a *contrast* to distinguish acute handling from chronic hypoxia. These are reported depth-matched, per nucleus, averaged across donors.

2.5 Anchor classification (cell-state composition)

Our central instrument reports related zonation-detection metrics on depth-normalized counts. **Pericentral (PC) anchors:** GLUL, CYP3A4. **Periportal (PP) anchors:** ASS1, PCK1, HAL, ALDOB. The headline PC fraction requires at least one PC UMI and fewer than two PP UMIs; the PP fraction requires at least two PP UMIs and no PC UMI; strict dual co-expression requires at least one PC marker at ≥ 2 UMIs and at least two PP markers at ≥ 2 UMIs each; and null means neither pole passes the permissive detection gate. Because these thresholds answer different questions, they are *overlapping measurements rather than an exhaustive four-class partition*; the remaining weak-dual fraction is reported explicitly. Each headline fraction is averaged over eight deterministic thinning draws. The unit of inference is the **donor**; per-donor fractions are summarised by stage. We also swept a sensitivity grid (GLUL-only / CYP3A4-only / both; k in $\{1, 2\}$; 2-of-4 / 3-of-4 on the PP side; ALDOB in/out; CPS1; strict-identity variants) and a marker-set ladder spanning $2 \rightarrow 1{,}637$ genes.

2.6 Equivalence / bounding

We report the donor-level difference and its 90% confidence interval for F1-vs-F4 (n=8 vs 4) and F1-vs-F3 (n=8 vs 12), then evaluate the pre-specified ±5- and ±10-percentage-point equivalence margins. Equivalence is claimed only when the full interval lies inside the chosen margin; the observed interval width is also reported directly as a precision result.

2.7 Pseudobulk differential expression

For "what else changes," we summed each biopsy donor's hepatocyte raw UMIs into one count profile (**donor-level pseudobulk**, the recommended guard against pseudoreplication; Squair et al. 2021) and ran **edgeR**: `filterByExpr` against the design ($\rightarrow$ 21,022 genes retained), TMM normalization (which corrects depth *and* composition, the proper fix for the per-10k-denominator problem), and a negative-binomial quasi-likelihood GLM with robust dispersion (common BCV ~ 0.405). The primary contrast is **F4-vs-F1**; a 5-level omnibus fibrosis test is a secondary screen, and interior contrasts (F2/F3-vs-F1) guard against the thin F4 (n=4). A housekeeping panel (ACTB, GAPDH, B2M, MALAT1) was tracked and stayed flat, confirming the normalization.

2.8 Ambient / compositional checks

Any DGE hit was audited for source: (i) **cross-lineage abundance**; (ii) **cholangiocyte fraction by stage**; (iii) per-cell co-expression of $\geq$2 biliary markers in hepatocyte-labeled nuclei; (iv) a saved **decontX sensitivity reanalysis** on the cellular count matrix followed by donor pseudobulk DGE; and (v) a crude **hepatocyte-cholangiocyte doublet probe** (KRT19/KRT7 $\geq$2 UMI), treated explicitly as exploratory. These checks probe compatibility with compositional explanations; none identifies the source on its own. The decontX result is scoped to the per-gene biliary analysis, not the CYP gene-set trend.

Choice	What	Why (custom, non-standard)
Counts	raw RNA UMIs	SCT is a model output; molecule-level co-detection/ambient questions need raw counts
Unit	donor ($\sim$47)	avoids pseudoreplication of $\sim 10^4$ nuclei (Squair et al. 2021)
Disease axis	needle biopsies, F1$\rightarrow$F4	removes the acquisition confound; endpoints are organ cubes
Depth control (classification)	binomial thinning to 1,500 UMIs, MC-averaged	equalizes *detection* across nuclei without a denominator
Anchor metrics	mixed explicit thresholds	PC, PP, strict dual, null and weak-dual answer related but non-exhaustive questions
Depth control (DGE)	TMM (keep all counts)	thinning would discard the power the count model needs
DGE model	edgeR NB-QL GLM, robust dispersion	small-n calibration (F4 n=4); robust QL handles thin strata
Hit audit	cross-lineage, decontX, doublet probe	probes compositional explanations without closing source attribution

Compact methods table.

3 Results

3.1 Tissue acquisition is a major confound, collinear with disease stage

The single most consequential fact about this cohort is structural, not statistical: **disease stage is collinear with how the tissue was obtained**. The F0–F4 disease spectrum is needle biopsy; both endpoints — healthy and end-stage — are deceased-donor organ cubes (Sec. 2.1, verified against Paper 1's Methods). So "healthy $\rightarrow$ MASLD $\rightarrow$ end-stage" is simultaneously "organ cube $\rightarrow$ needle core $\rightarrow$ explant." Because every milder-disease sample is a biopsy and every end-stage sample is an explant,

disease and acquisition cannot be separated in the pooled comparison. Sequencing batch compounds this: run almost perfectly predicts tissue source (Cramér's V(run, source) = 0.84, near the maximum of 1; 9 of 13 runs carry a single stage), so batch and source are not independently estimable at the endpoints. The methodological response is forced: exclude the procurement-confounded endpoints from the disease axis and re-test on the acquisition-matched biopsies. One sub-confound was separately cleared — within explants the zonation *detection* pattern is lobe-invariant (e.g. GLUL detection 0.350/0.343/0.297 across right/caudate/left; ALDOB 0.821/0.837/0.838), so multi-lobe/caudate sampling does not manufacture the signal — but this clears only the lobe sub-confound, not the larger explant-vs-biopsy one.

3.2 The endpoint groups carry a cross-lineage stress signature

The endpoints are not just acquired differently; they carry a measurable stress signature. Depth-matched stress-module UMIs per nucleus (mean across donors) are **0.074 in needle biopsies, 0.282 in healthy organ cubes (∼3.8×), and 1.675 in end-stage explants (∼22.6× the biopsy level)**; the ambient-robust ≥2-UMI detection rate follows the same ladder (0.053 / 0.247 / 0.770). The explant-vs-biopsy fold is similar in hepatocytes (immediate-early genes ∼18.5×) and in **endothelial cells (∼18.2×)**, a lineage without a hepatocyte zonation program. Heat-shock chaperones rise ∼66× in hepatocytes, while the HIF/hypoxia contrast is only ∼1.7–2.6×. This cross-lineage pattern is compatible with acute acquisition or handling effects and supports treating tissue source as a serious alternative explanation; it does not identify procurement as the sole cause.

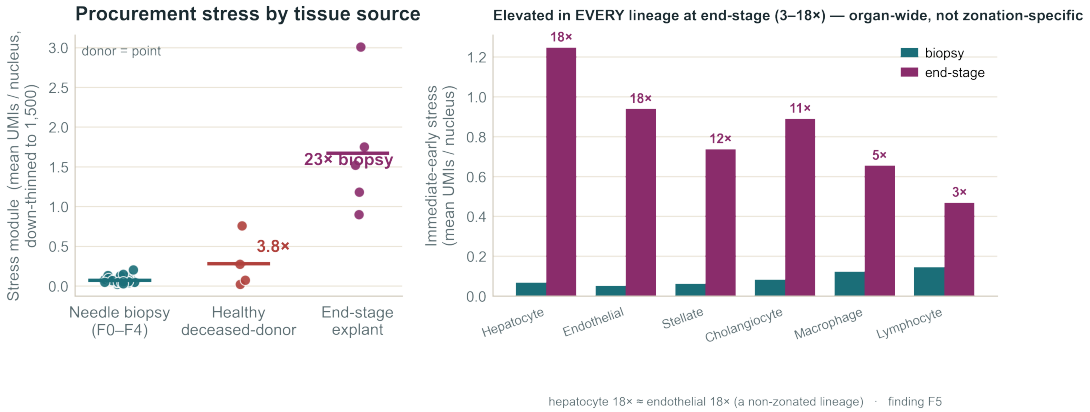


Figure 1: Stress-module signal by tissue source and across lineages. Depth-matched immediate-early and heat-shock UMIs per nucleus are low across needle biopsies and rise in the deceased-donor organ-cube endpoints. The similar explant-vs-biopsy fold in hepatocytes and endothelial cells supports an organ-wide acquisition or handling contribution, without identifying its sole cause.

3.3 Matched-biopsy measurements show no clear progressive de-zonation signal

On the acquisition-matched biopsy axis, the selected zonation measurements are flat or non-monotone. Across F0 through F4, median pericentral-anchor fractions are 33.86, 18.24, 23.70, 19.20 and 23.15%; periportal-anchor fractions are 19.33, 19.58, 22.27, 23.89 and 23.12%; and null fractions are 36.10, 44.44, 35.70, 39.51 and 40.17% (F0 rests on n=2 and is descriptive). Strict dual co-expression is rare at 0.18–0.36%, while 6.87–10.00% of nuclei fall into the explicitly reported weak-dual remainder created by the mixed thresholds. A broad PC↔PP gradient is present at every biopsy stage, including cirrhotic F4. In short, these measurements show **no clear progressive transcriptional de-zonation signal** on the matched axis; this is not a positive demonstration that zonation is preserved.

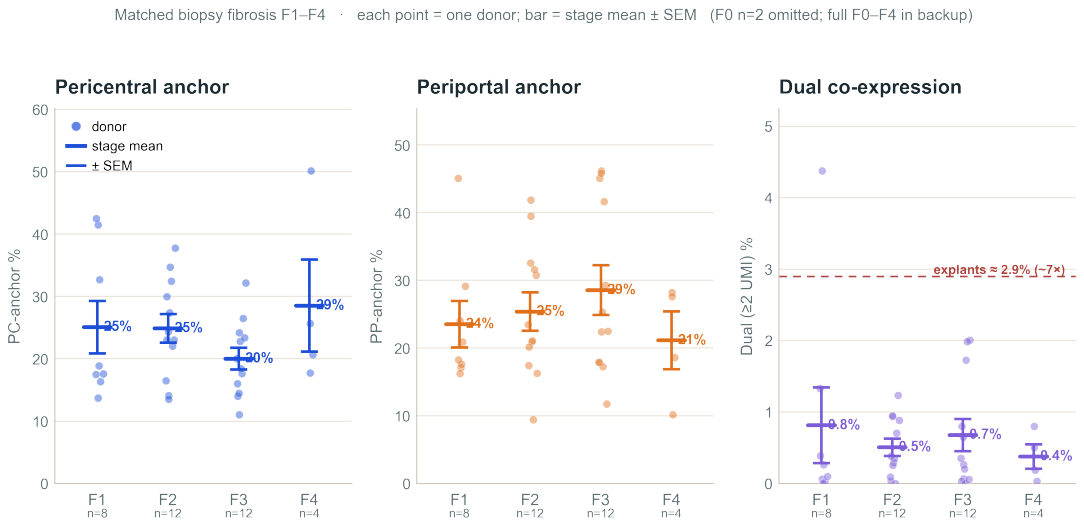


Figure 2: Matched-biopsy anchor classification (3-panel). Pericentral-anchor, periportal-anchor and strict (≥2-UMI) dual co-expression fractions across needle-biopsy fibrosis F1–F4 (box = per-donor distribution, each point = one donor; F0 n=2 omitted, in backup). The pericentral and periportal fractions are flat/non-monotone; the dual fraction is near-zero (∼0.4%, vs ∼2.9% in explants) and shows no trend. Null and PP:PC are in Figure 3.

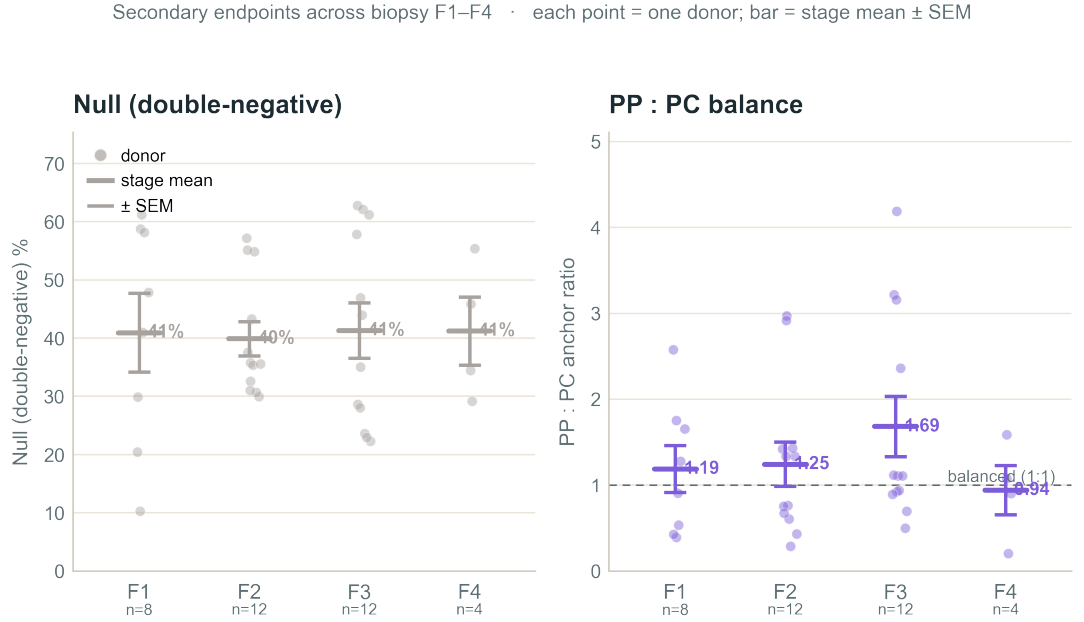


Figure 3: Secondary endpoints (2-panel). Null (double-negative) fraction and the periportal:pericentral anchor ratio across biopsy F1–F4 (box = per-donor distribution, each point = one donor). Null ≈44/36/39/39% (no rise → no turn-off); PP:PC ≈1.16/1.01/1.10/1.18, hovering around the balanced 1:1 line — both flat/non-monotone.

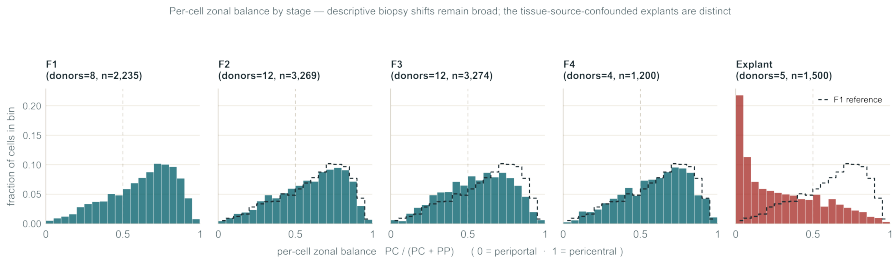


Figure 4: Per-cell zonal balance by stage (down-thinned to 1,500 UMIs, ≤300 nuclei/donor). The x-axis is each nucleus's pericentral share PC/(PC+PP): 0 = periportal, 1 = pericentral. The dashed outline repeats the F1 distribution as a descriptive reference. Biopsy-stage distributions remain broad with non-monotone shifts; the explant distribution is distinct but is confounded with tissue source.

3.4 The pattern is robust; precision remains limited

The flat/non-monotone pattern is not a single fragile contrast: it recurs across the sensitivity grid (anchor choice, UMI threshold, marker-set ladder from 2 to 1,637 genes), and the headline metrics are averaged over eight deterministic thinning draws. The donor-level F1-to-F4 pericentral-fraction difference is **+0.0345 (+3.45 percentage points), 90% CI −0.1363 to +0.2054**; for F1-to-F3 it is −0.0505, 90% CI −0.1334 to +0.0324. The full intervals do not lie inside either the pre-specified ±5- or ±10-point margins, so **equivalence is not established**. The result is therefore descriptive and precision-bounded: no clear progressive shift is observed, but modest shifts remain compatible with the donor-level uncertainty. The claim applies to transcriptional measurements, not lobule-scale spatial architecture, which snRNA cannot see.

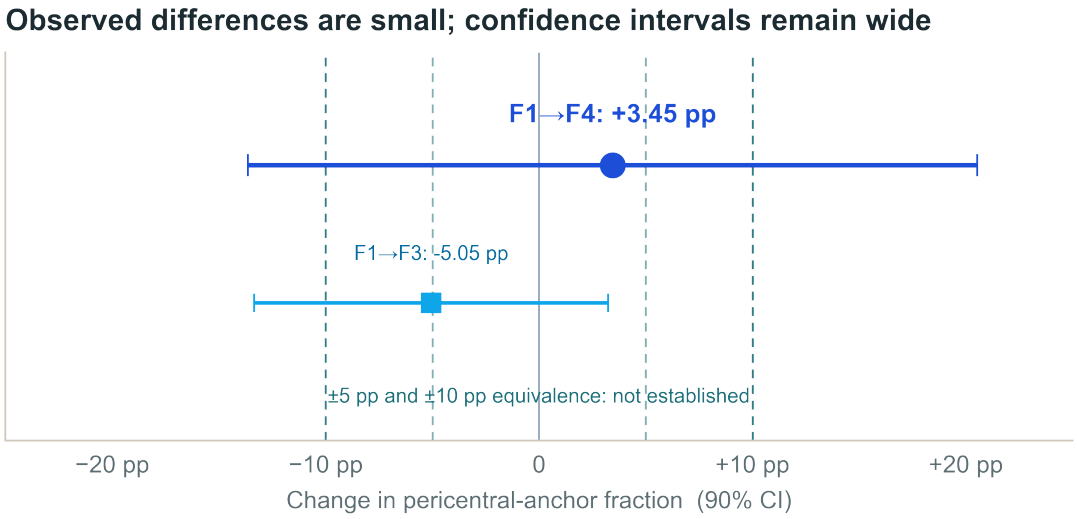


Figure 5: Precision and fixed-margin equivalence checks for the donor-level pericentral-anchor fraction. The 90% confidence intervals for F1-to-F4 and F1-to-F3 cross the pre-specified ±5- and ±10-point margins; neither equivalence claim is established.

3.5 End-stage explants are heterogeneous, not one program

Setting the explants aside as confounded does not mean they are uninformative about *why* pooling fakes a collapse. The five explants go in opposite directions rather than sharing one end-stage phenotype: **CL104** retains its pericentral pole (PC-anchor 49.7%, PP:PC 0.13); **CL16** nearly collapses to periportal (PC 3.2%, PP:PC 20.3); **CL18** floods into co-expression (dual 22.4%); CL103 and CL17 are periportal-leaning. Across the five, the pericentral-anchor fraction spans 3–50% and PP:PC spans 0.13–20. Pooling five discordant organs into one marker correlation manufactures a single uniform "collapse" that none of them individually shows — the signature of heterogeneous, separately-handled

organ procurement.

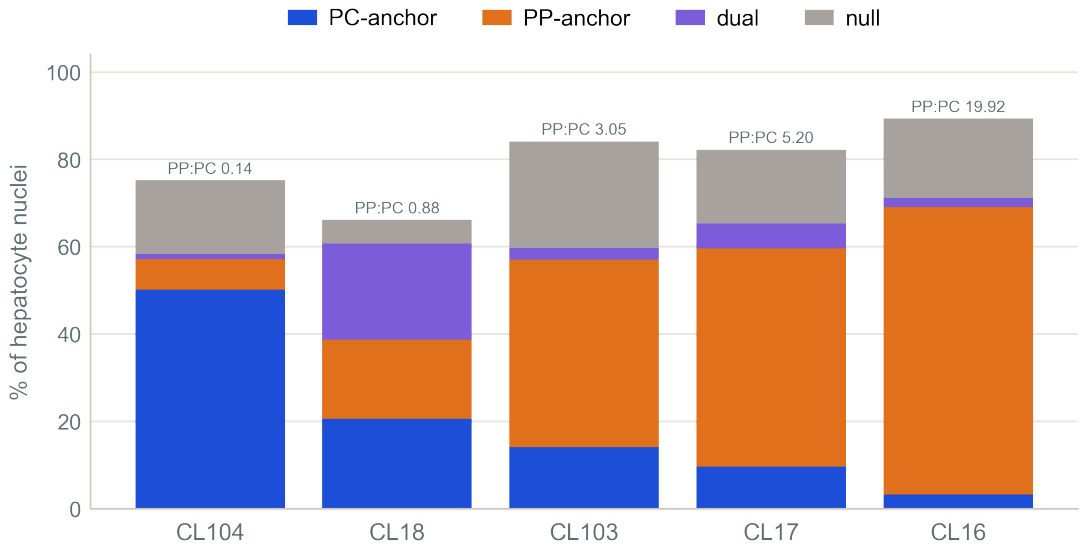


Figure 6: End-stage explant heterogeneity. Per-explant anchor profiles: the five organs span retained pericentral identity (CL104) to near-total periportal collapse (CL16) to a co-expression flood (CL18). A pooled correlation collapses these into one number.

3.6 Genome-wide per-gene DGE finds a biliary signal, not zonation loss

A genome-wide donor-level pseudobulk DGE scan plays two roles. As an **independent count-based check** on the zonation result, the zonation genes are flat at the expression level — GLUL F4-vs-F1 FDR **0.80** (i.e. far above the 0.05 significance threshold, so no detectable change in the canonical pericentral identity gene), CYP3A4 0.85, ASS1 0.96, with all detox/periportal markers FDR > 0.79 — agreeing with the anchor classification (this is a corroborating sanity check, *not* the primary proof). As a **discovery scan**, of the 21,022 tested genes only **64 reach FDR<0.05 at F4-vs-F1** (omnibus across stages: 91), and they form a coherent **biliary/ductular marker set**: EPCAM (logFC +2.3, FDR 0.002), GRHL2 (+3.1, FDR 0.0004), SPINT2 (+2.7, FDR 0.003), SOX4 (+2.5, FDR 0.047), SOX9 (+2.4, FDR 0.044), B3GNT3 (+2.9, FDR 0.001), plus CXCL10 (+5.4, FDR 0.001). The honest claim is bounded: **no large single-gene hepatocyte program changes across the matched axis other than this biliary rise** — but a per-gene scan can miss a weak *coordinated* program. Gene-set and per-cell tests in Sec. 3.7 support one such biopsy-internal trend. A within-class confirmatory DGE (Plan B) found the same biliary set appearing class-agnostically across PC/PP/null/dual nuclei, with zonation genes flat inside every class.

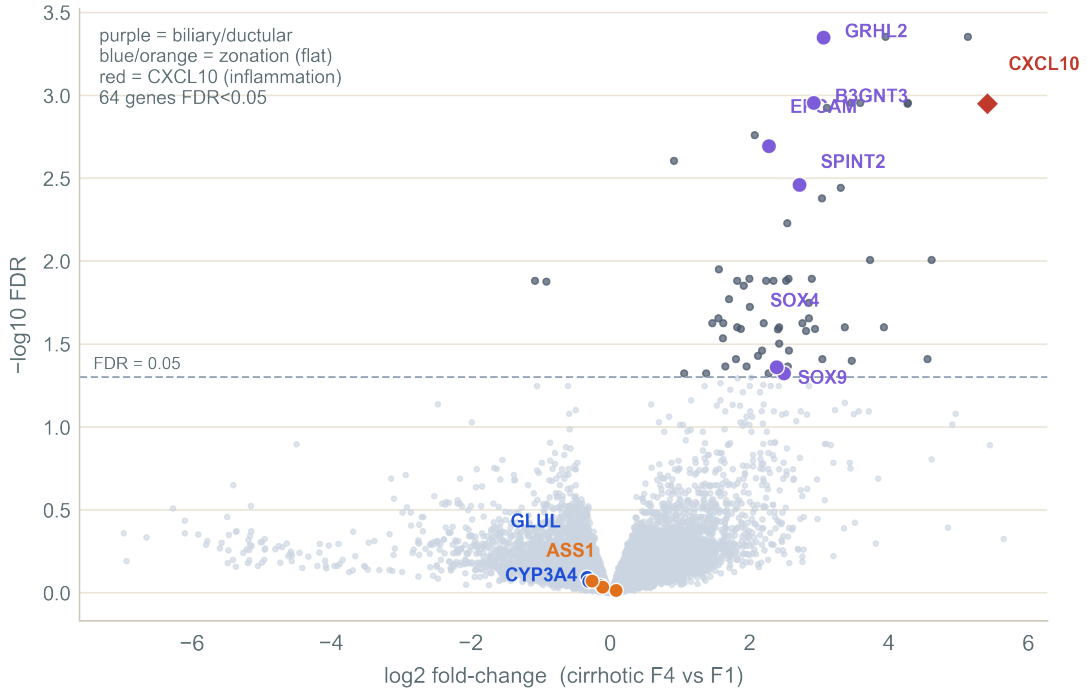


Figure 7: Genome-wide pseudobulk DGE (F4-vs-F1) volcano. Zonation genes sit at the origin (flat; GLUL FDR 0.80). The significant hits are a biliary/ductular marker set (EPCAM, GRHL2, SPINT2, SOX4, SOX9, B3GNT3) plus CXCL10.

3.7 The one biopsy-internal program that does change: the pericentral detox program dims

The per-gene scan above is blind to a program spread thinly across many genes. To probe for one we ran two complementary, **pre-specified** gene-set analyses on the same donor-level pseudobulk, plus a depth-matched per-cell measure that reduces, but cannot eliminate, compositional sensitivity. The ten sets were fixed in advance from CYP/detox biology and Paper 2's own landmark genes (pericentral identity, periportal identity, phase-I CYP detox, phase-II conjugation, urea cycle, bile-acid/lipid, cholangiocyte/ductular) with three biological controls (interferon, EMT/fibrogenesis, ER-stress).

Gene-set tests. Using `limma::camera` (competitive: is the set more shifted than the rest of the transcriptome?) and `mroast` (self-contained: does the set move against zero?), with GSEA pre-rank as a third, independent read, the directions agree across all three. The **pericentral detox programs go down** with fibrosis: phase-I CYP detox (camera FDR 2×10^{-6}, GSEA NES -2.04), the pericentral landmark set (Paper 2 landmarks; FDR 1.5×10^{-4}, NES -1.88), and phase-II conjugation (FDR 7×10^{-3}). The biological controls behaved exactly as biology demands — EMT/fibrogenesis up (FDR 9×10^{-6}), interferon/inflammation up (FDR 1.5×10^{-3}), ER-stress flat (FDR 0.72, not significant) — which is what licenses reading the detox-down call as real rather than an artifact. The biliary/ductular set rises (1.9×10^{-5}), the periportal-anchor set rises mildly, and urea-cycle and bile-acid/lipid do not move. The simultaneous pericentral-down / periportal-up pattern is the see-saw a *competitive* test on relative (CPM) data would show if rising biliary/stromal genes simply ate the library budget; we therefore controlled for it adversarially.

Adversarial verification. Four checks reduce the chance that the gene-set result is a normalization artifact. (i) Re-running camera on **TMM**-normalized counts leaves the detox FDRs essentially unchanged (CYP $2.3 \times 10^{-6} \rightarrow 3.5 \times 10^{-6}$). (ii) The **self-contained** `mroast` test still clears the CYP detox core (Down FDR 0.015); the broader pericentral set is only borderline (0.08), so the strictly self-contained evidence is carried by the CYP core. (iii) Dropping the 65 up-going genes and renormalizing leaves detox Down with unchanged FDRs. (iv) An independently computed, depth-matched

measure — detox transcripts per pericentral nucleus after down-thinning every nucleus to 1,500 UMIs — gives F0–F4 means of 12.48/11.86/10.21/10.04/9.02 transcripts per nucleus when averaged over eight deterministic draws, with donor-level Spearman $\rho = -0.461$ ($p = 0.0041$). We lead with this fibrosis-wide trend rather than the F1-vs-F4 endpoint, because F4 has only four biopsy donors.

Interpretation boundary. The robust change is a **relative decline in the CYP xenobiotic program**: this is the only set that clears the self-contained `mroast` test on its own (FDR 0.015), and it is accompanied by the repeat-averaged per-cell trend. The broader pericentral landmark and phase-II sets reach the competitive bar but fail the self-contained test (roast FDR 0.081 and 0.220), no single gene is individually significant, and the landmark set overlaps the detox genes. GLUL and CYP3A4 are individually flat (per-gene F4-vs-F1 FDR 0.80 and 0.85), while the selected anchor measurements show no clear progressive redistribution. Three caveats are load-bearing: the decline is **relative** within a fixed 1,500-UMI budget, not proven absolute molecule loss; it is a **donor-level trend**; and it could reflect a shift among pericentral subzones rather than pure within-cell downregulation. We therefore describe a dimming transcriptional program without claiming preserved cell identity or a specific cellular mechanism.

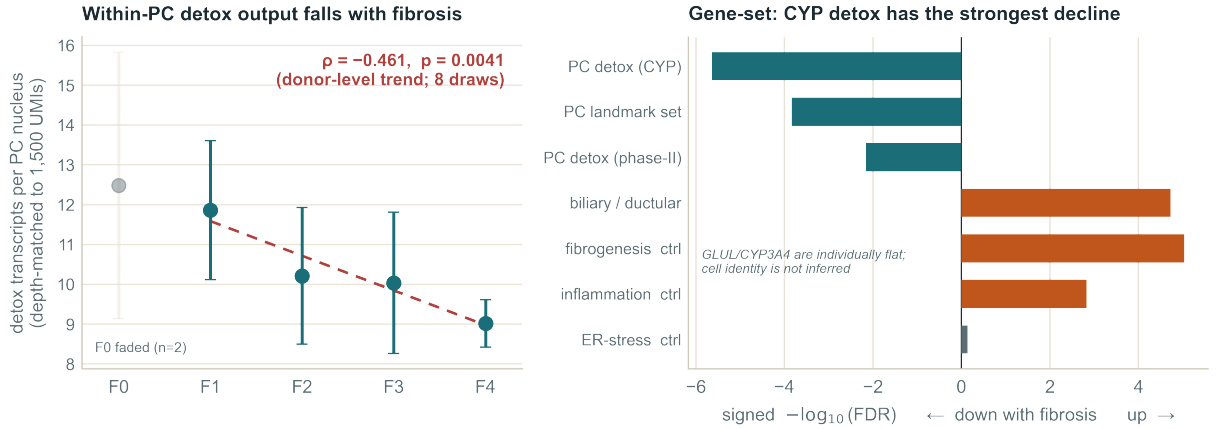


Figure 8: The pericentral detox program declines in relative abundance. *Left:* within-PC detox output, depth-matched to 1,500 UMIs and averaged over eight deterministic draws, falls from 11.86 (F1) to 9.02 (F4); donor-level Spearman $\rho = -0.461$, $p = 0.0041$ (F0 faded, n=2). *Right:* pre-specified gene-set programs (`camera`, signed $-\log_{10}$FDR): the pericentral CYP detox set is the only set also clearing the self-contained test; the pericentral landmark and phase-II sets are competitive-only. GLUL/CYP3A4 are individually flat (FDR 0.80, 0.85).

3.8 The biliary signal has unresolved compositional and intrinsic explanations

Several observations are compatible with a compositional contribution, but none closes the source question. (i) Every headline biliary gene is **5–78× more abundant in cholangiocytes than in hepatocytes**. (ii) The cholangiocyte fraction rises to **∼8% at F4**, so ductular reaction increases the opportunity for ambient or co-capture effects. (iii) Only **∼0.4%** of hepatocyte-labelled nuclei robustly co-express ≥2 biliary markers, arguing against a widespread state while leaving a rare subpopulation unresolved. (iv) In the saved decontX sensitivity output, the hit count falls from 64 to 34 and **SOX4 (FDR 0.131) and SOX9 (FDR 0.118)** move below significance, while **EPCAM, B3GNT3, SPINT2 and CXCL10 survive**. That optional heavy analysis was not rerun in the bounded verified route, and decontX can under- or over-correct; this result is therefore a sensitivity check, not a source label. A crude doublet probe rises with fibrosis and has the inflated-total-count pattern expected from co-capture, but it cannot separate doublets from rare biphenotypic or transdifferentiating cells. **CXCL10 is considered separately**: its hepatocyte expression does not track the cholangiocyte fraction (correlation −0.09), making it a candidate inflammatory signal. The bounded conclusion is that ambient RNA, co-capture and a smaller intrinsic component all remain plausible; this cohort does not quantify their relative contributions.

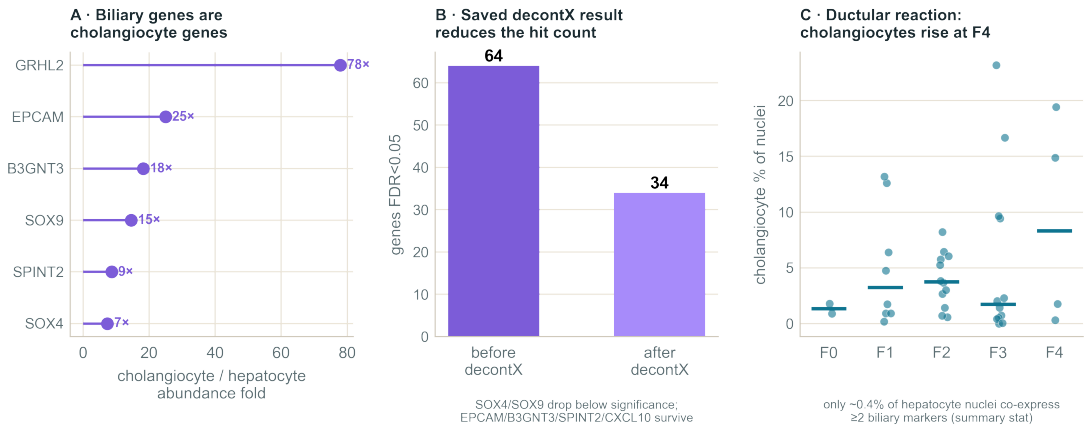


Figure 9: Source-attribution probes for the biliary signal. Left: headline biliary genes are 5–78× cholangiocyte-dominant across lineages. Middle: the saved decontX sensitivity output reduces hits from 64 to 34. Right: the cholangiocyte fraction rises into F4. The combined evidence is compatible with a compositional contribution but does not distinguish ambient RNA, co-capture and a rare intrinsic state.

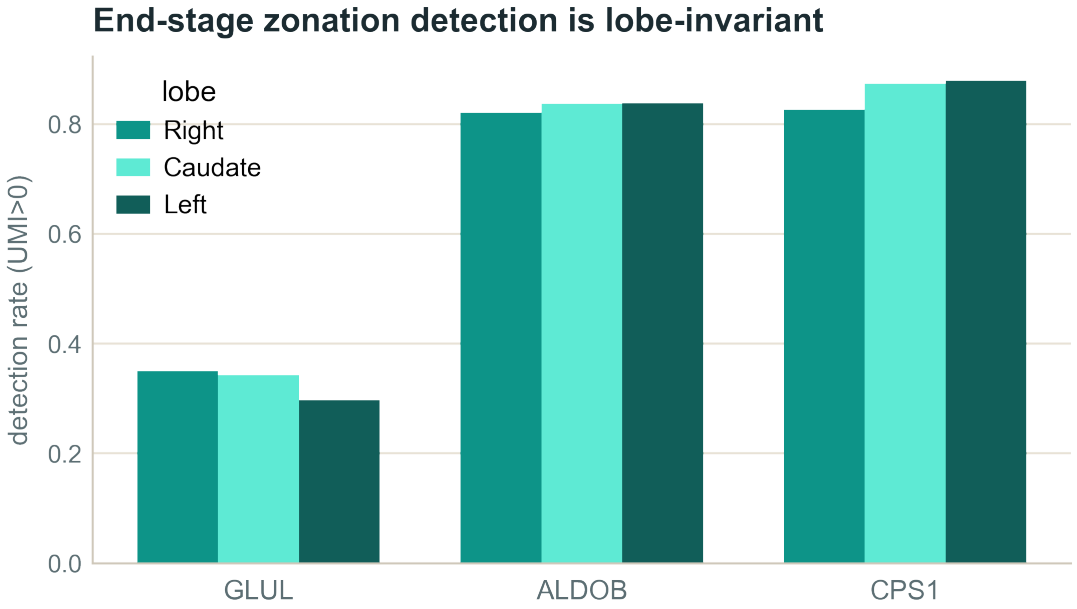


Figure 10: Lobe invariance (supporting). Within explants, the zonation-marker detection pattern is invariant across right/caudate/left lobes (e.g. GLUL 0.350/0.343/0.297), so multi-lobe sampling does not manufacture the de-zonation signal — clearing one sub-confound.

Two conceptual figures belong with this report but are not fabricated here. A **study-design / confound schematic** would show the disease axis (F0–F4 needle biopsies, right lobe) beside the end-points (healthy and end-stage deceased-donor organ cubes, multi-lobe), with the stage-vs-acquisition collinearity stated explicitly: because every milder sample is a biopsy and every end-stage sample is an explant, disease stage and acquisition method are perfectly collinear and not separable in the pooled comparison *(schematic; see presentation slides 3 and 5)*. An **anchor-method pipeline schematic** would show raw counts → binomial down-thinning to 1,500 UMIs → ≥2-UMI strict anchor calls → PC/PP/dual/null per nucleus → donor-level fractions → sensitivity grid *(schematic; see presentation slides 5 and 8)*.

4 Discussion

4.1 Summary

Four results, at decreasing confidence levels. **Main result:** across acquisition-matched needle-biopsy MASLD, selected zonation measurements show **no clear progressive transcriptional de-zonation signal**; however, the 90% confidence intervals do not fit inside the pre-specified ±5- or ±10-point margins, so equivalence is not established. The strong endpoint pattern is confounded with tissue acquisition, and the cross-lineage stress signal is compatible with an acquisition or handling contribution. **Confirmed biopsy-internal trend:** the pericentral *detox* program declines in relative abundance across fibrosis. The CYP xenobiotic core is significant in both a competitive (camera FDR 2×10^{-6}) and a self-contained (roast FDR 0.015) gene-set test, and the eight-draw per-cell measure declines from 11.86 to 9.02 transcripts per nucleus from F1 to F4 (Spearman $\rho = -0.461$, $p = 0.0041$). The broader landmark and phase-II sets are competitive-significant but not self-contained, and no single gene is significant. This result does not prove absolute molecule loss or distinguish within-cell regulation from a shift among pericentral subzones. **Secondary result:** the one per-gene genome-wide change is a biliary/ductular marker rise (64 hits) inside hepatocyte-labeled pseudobulk; zonation genes are flat (GLUL FDR 0.80). **Interpretive lead:** the source of that biliary rise remains unresolved; ambient RNA, mixed/doublet nuclei and a small intrinsic contribution remain plausible.

4.2 Meaning for liver biology

At the level snRNA can measure, the biopsy donors retain a broad pericentral↔periportal gradient and the selected redistribution measurements do not change progressively across fibrosis. The confidence intervals remain wide enough that modest shifts cannot be ruled out. Within nuclei meeting the pericentral gate, the detox program declines in relative abundance as fibrosis advances. These observations separate two measurements — balance and program magnitude — without proving that the same biological cells retain identity over time. The strong endpoint signal remains an end-stage/explant phenomenon confounded with tissue source, and the biliary burden may reflect ductular reaction, ambient RNA, mixed nuclei, or a smaller intrinsic component that this cohort cannot quantify.

4.3 Relation to the original paper

This is a correction of one evidence leg, not an overturning. We re-examine only the **single-nucleus transcriptional** de-zonation statistic — a relative marker correlation pooled across a sampling discontinuity — and find that, on acquisition-matched biopsy tissue and raw counts, it does not support *progressive, disease-driven* de-zonation. We explicitly do **not** touch the paper's immunofluorescence (GLUL/ASS1 co-localization), its whole-organ FLASH 3D imaging of end-stage architecture, or its organoid / PI3K-AKT-mTOR plasticity evidence; those measure different molecules and timescales than the acute RNA signal we analyse, and cirrhosis genuinely remodels the lobule. We also agree with the paper that the strong signal is an end-stage phenomenon — which is precisely the tissue we set aside as confounded. The disagreement is narrow: the matched-biopsy snRNA data do not show a clear progressive de-zonation signal, and the cellular source of the modest biliary rise remains unresolved.

4.4 Limitations

The matched axis is small at its extremes (F0 n=2, F4 biopsy n=4); neither the ±5- nor ±10-point equivalence margin is established. snRNA measures per-cell transcriptional balance, **not lobule geometry**; needle cores miss septa and perivenular zones, so our claim is limited to the absence of a clear progressive transcriptional de-zonation *signal*, not spatial or architectural preservation. Sequencing batch is not randomized (V(run, source)=0.84); within the biopsy axis the biliary hits survive a run covariate and a single-run test, but the design is not clean. The saved decontX sensitivity result was not rerun in the bounded verified route and must not be extended to the CYP gene-set

trend. The doublet probe is exploratory (n=16 suspects) and cannot separate a doublet from a rare transdifferentiating cell. The detox result is a **relative** decline within a fixed sequencing budget, a donor-level trend, and could partly reflect a shift among pericentral subzones rather than pure within-cell downregulation. This is a single cohort with no independent biopsy series and no protein or spatial validation of our own.

4.5 Challenges and lessons learned

The project's own progression record (archived deck and primer) documents a real methodological turn. The original instrument was a **relative, z-scored zonation "ruler"** built from a healthy spatial atlas and applied via marker correlation — and it produced the apparent collapse. Two problems retired it. First, it was **source-confounded**: pooled across the needle-biopsy↔organ-cube discontinuity, it manufactured a between-group "collapse" that dissolved within groups — a textbook Simpson's-paradox reversal (each healthy donor individually anti-correlated, yet pooling the four flipped the sign). Second, it was **intrinsically depth-sensitive and mechanism-blind**: turn-off, co-expression and pure noise all shrink a z-scored spread identically, so the ruler could not say *how* zonation was changing. The fixes followed directly: move to **raw molecule counts** (the shipped matrix is an SCT output, unusable for co-detection and ambient sensitivity); equalize sequencing depth by **binomial down-thinning** so detection differences reflect biology; classify nuclei by **strict ≥2-UMI anchors** that separate the failure mechanisms the ruler conflated; and make the **donor the unit of inference** to avoid pseudoreplicating tens of thousands of nuclei (one F4 donor alone contributes ∼59% of F4 cells). The hardest lesson was structural rather than technical: the confound discovery — that disease stage is collinear with tissue acquisition — reframed the entire project from "measure how much zonation degrades" to "test whether the degradation survives a matched-source comparison," and it is what carries the conclusion, no causal claim about stress switching off GLUL required.

4.6 Future work

The owed **gene-set / pathway test** — CAMERA (competitive) plus ROAST/mroast (self-contained), with GSEA pre-rank — has now been run over pre-specified hepatocyte programs (Sec. 3.7) and is what surfaced the pericentral-detox dimming; the highest-priority next step is therefore to **validate that dimming gradient** — in spatial / independent-biopsy tissue, and with a functional readout of pericentral detox capacity — and to settle whether it is within-cell downregulation or a pericentral-subzone shift. Beyond that: a **leave-one-F4-donor-out** DGE to test robustness of the biliary hits to the thin F4 stratum; a **quantitative contamination model** to ask whether the estimated ambient/doublet load quantitatively accounts for the biliary counts; **spatial validation** on biopsy-matched tissue; and replication in an **independent biopsy cohort** with clinical covariates.

The transferable lesson is the reflective one. Relative summary statistics — marker correlations, spreads, z-scores — applied to cohorts where disease stage tracks tissue acquisition can manufacture biology that is not there. Count-based, depth-matched, donor-level analysis with an explicit confound audit is the more conservative standard, and it is what turned an apparent progressive collapse into a bounded, honest null with a single clearly-scoped lead left open.

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All entries web-verified against the primary sources (authors, title, journal, year, volume/pages).

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6 Figures and tables

#	Item	Source file
Fig 1	Procurement stress by source + cross-lineage	`fig_stress.png`
Fig 2	Matched-biopsy anchors — PC / PP / dual (3-panel)	`fig_result3.png`
Fig 3	Secondary endpoints — null + PP:PC (2-panel)	`fig_secondary.png`
Fig 4	Per-cell zonal balance by stage (F1 dashed reference)	`fig_gradient.png`
Fig 5	Precision and fixed-margin equivalence checks	`fig_tost.png`
Fig 6	End-stage explant heterogeneity	`fig_explant.png`
Fig 7	Genome-wide per-gene DGE volcano	`fig_volcano.png`
Fig 8	Pericentral-detox dimming + gene-set programs	`fig_dimming.png`
Fig 9	Source attribution (3-panel)	`fig_compositional.png`
Fig 10	Lobe invariance	`fig_lobe.png`
—	Study-design / confound schematic	*(schematic; see slides 3 and 5)*
—	Anchor-method pipeline schematic	*(schematic; see slides 5 and 8)*